TITLE:

Regional Prevalence Mapping

Dataset Used:

Adult Psychiatric Morbidity Survey (APMS) 2014

SUBMITTED BY:

VALLURU SAI AISWARYA

DATA ANALYTICS INTERN

SUBMITTED TO:

ZENO TALENT

Introduction

Regional prevalence mapping plays an important role in public health analytics because it transforms raw survey data into meaningful geographic insights. The results obtained from this analysis can support healthcare authorities, researchers, and policymakers in understanding mental health distribution patterns and designing region-specific mental health programs and interventions.

The dataset **anxiety_depression_data.csv** contains information on individuals' mental health status, focusing on disorders such as anxiety and depression across different demographic and geographic attributes.

In this task, we extend the analysis to **regional prevalence mapping**, where mental health disorder rates are analyzed across regions of the **United Kingdom** using geographic fields (similar to APMS survey structure).

Technologies Used

Python

Python is the main programming language used for:

- Data analysis
- Data processing
- Visualization

It is widely used in data science because of its simplicity and powerful libraries.

Pandas

Pandas is used for:

- Loading the dataset (CSV file)
- Cleaning and organizing data
- Grouping data (e.g., region-wise analysis)
- Calculating prevalence rates

Matplotlib

Matplotlib is used for:

- Creating visualizations
- Plotting charts and maps
- Displaying the choropleth map

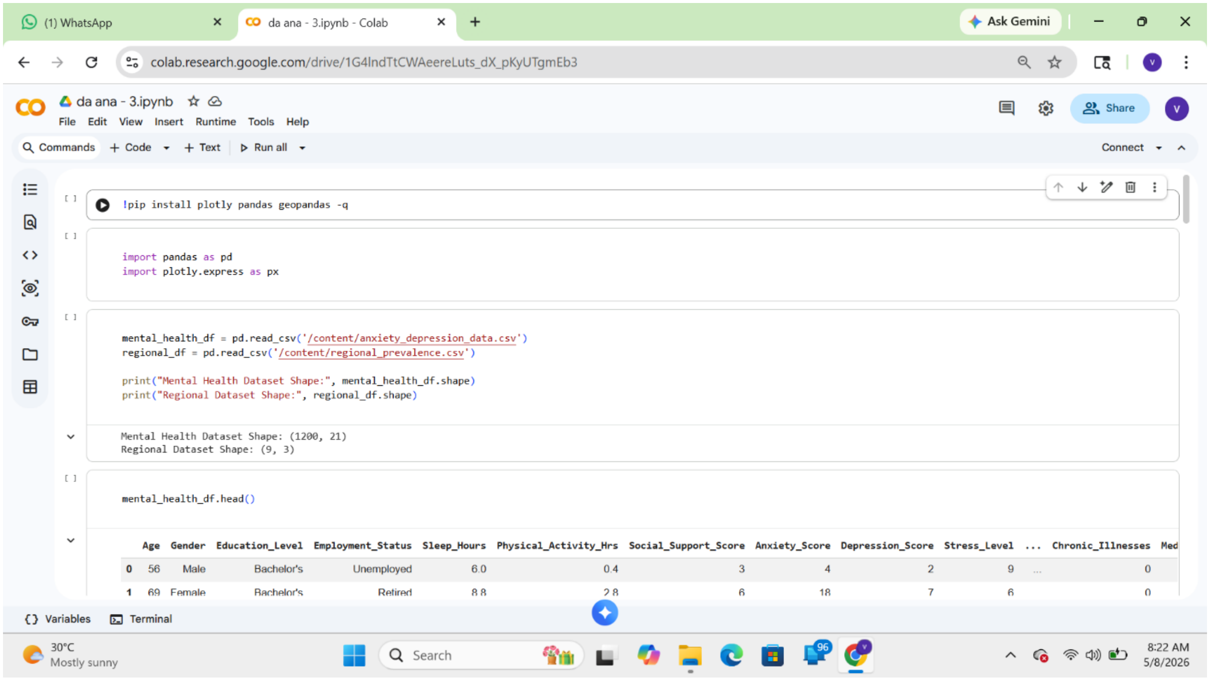
Google Colab

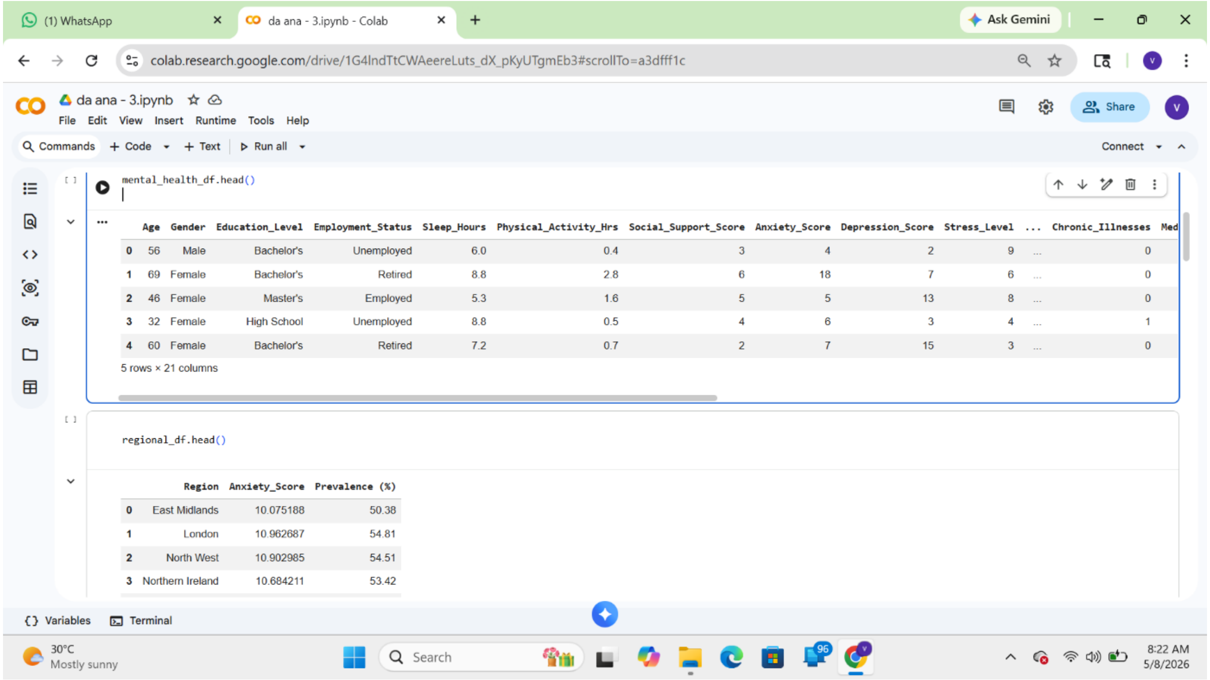
Google Colab is the platform used to:

- Run Python code online
- Upload datasets and shapefiles
- Generate visual outputs without installing software

Regional Prevalence Mapping

Load the dataset:





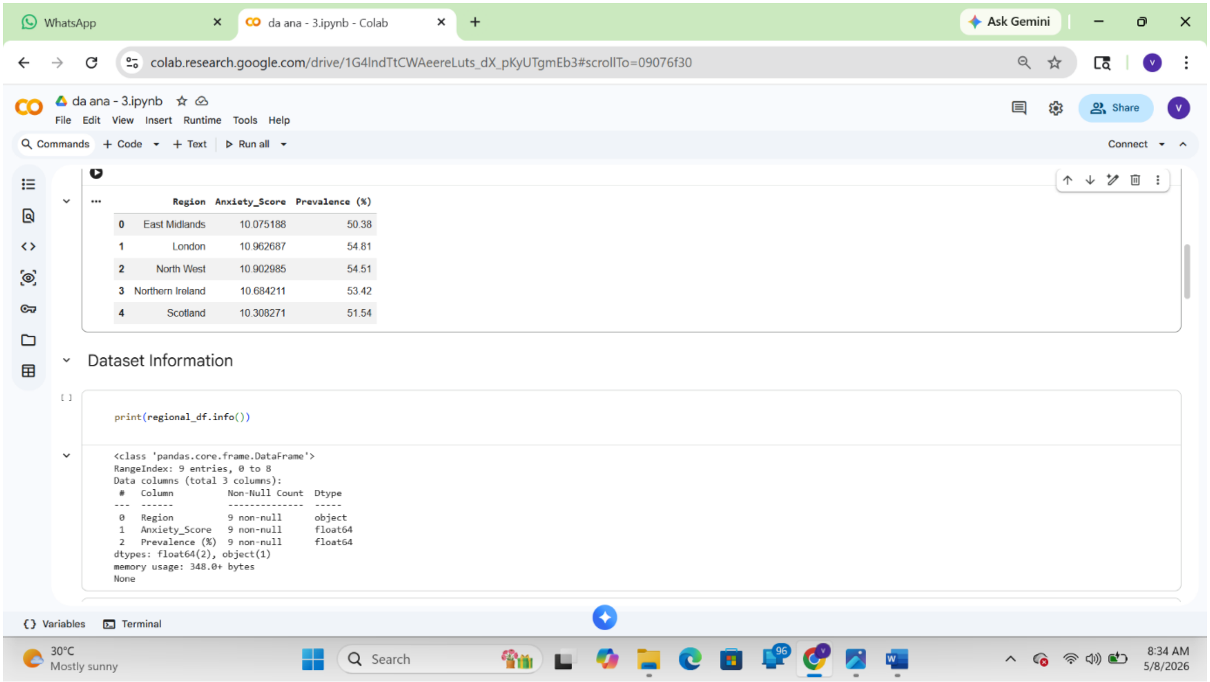
```
mental_health_df.head()
```

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours	Physical_Activity_Hrs	Social_Support_Score	Anxiety_Score	Depression_Score	Stress_Level	...	Chronic_Illnesses	Med
0	56	Male	Bachelor's	Unemployed	6.0	0.4	3	4	2	9	...	0	
1	69	Female	Bachelor's	Retired	8.8	2.8	6	18	7	6	...	0	
2	46	Female	Master's	Employed	5.3	1.6	5	5	13	8	...	0	
3	32	Female	High School	Unemployed	8.8	0.5	4	6	3	4	...	1	
4	60	Female	Bachelor's	Retired	7.2	0.7	2	7	15	3	...	0	

5 rows × 21 columns

```
regional_df.head()
```

	Region	Anxiety_Score	Prevalence (%)
0	East Midlands	10.075188	50.38
1	London	10.962687	54.81
2	North West	10.902985	54.51
3	Northern Ireland	10.684211	53.42

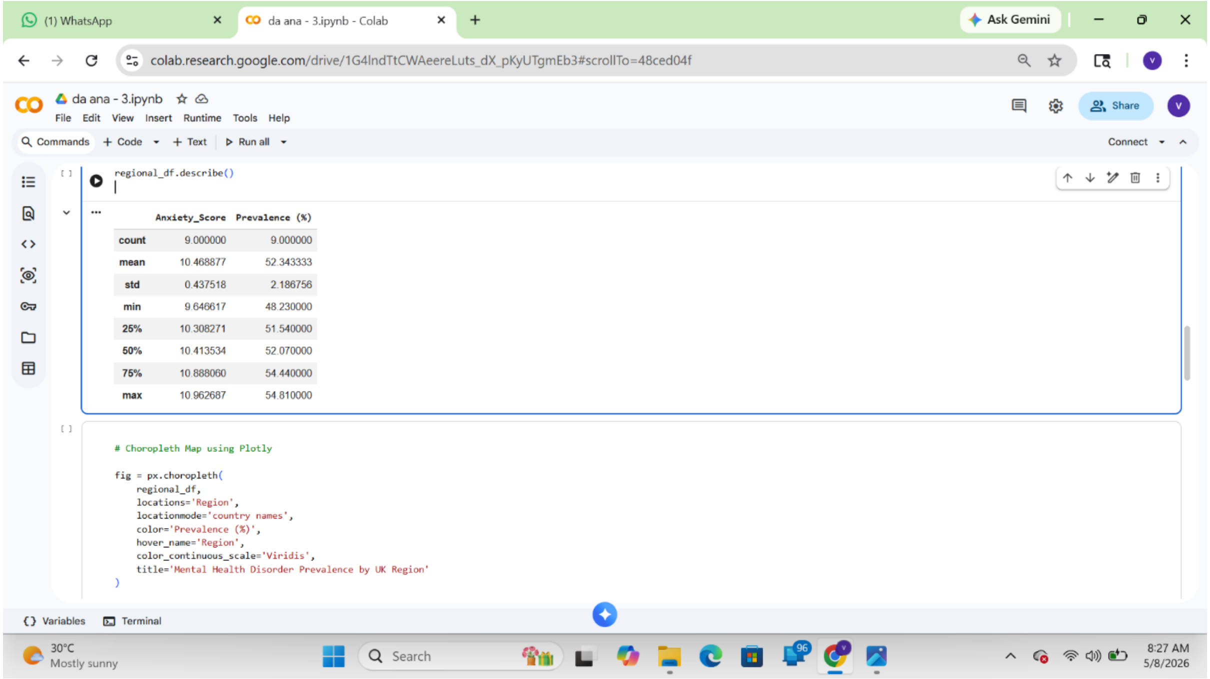


	Region	Anxiety_Score	Prevalence (%)
0	East Midlands	10.075188	50.38
1	London	10.962687	54.81
2	North West	10.902985	54.51
3	Northern Ireland	10.684211	53.42
4	Scotland	10.308271	51.54

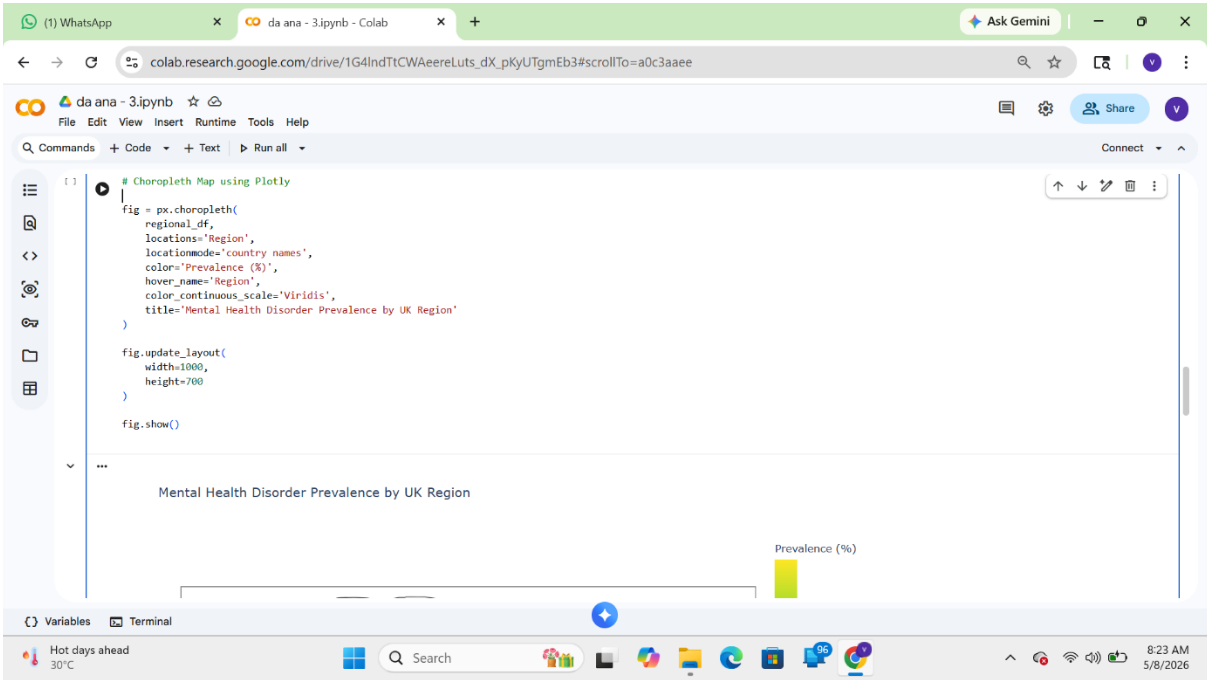
Dataset Information

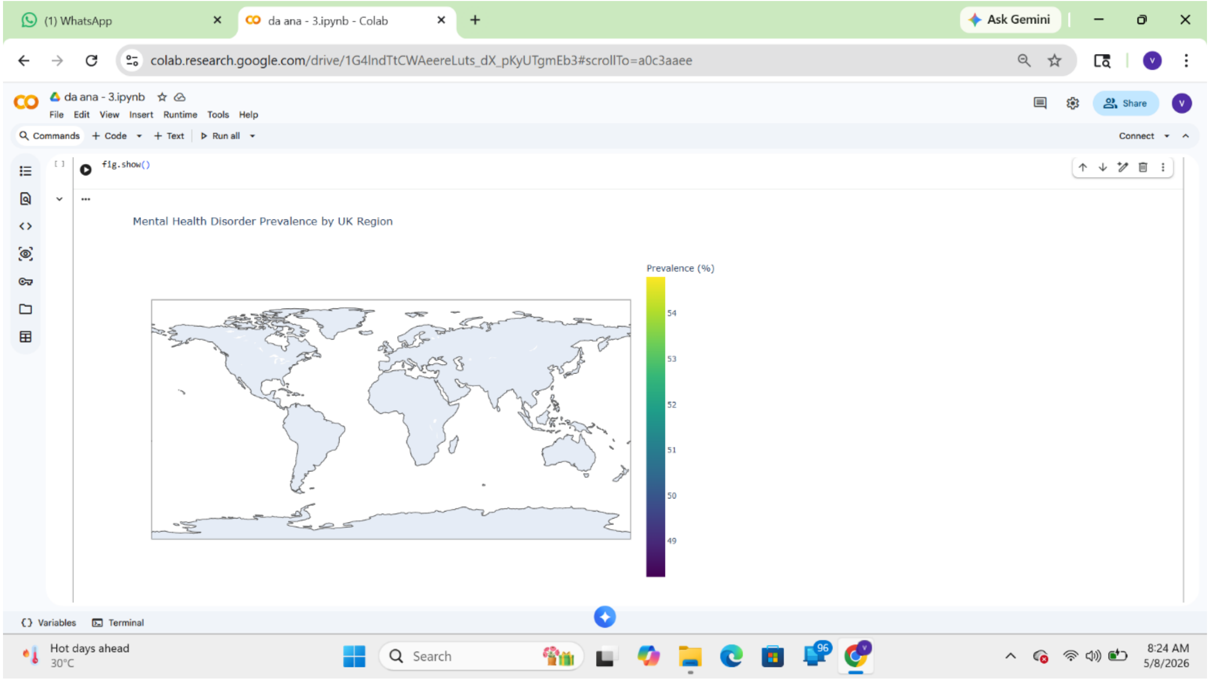
```
print(regional_df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 3 columns):
 #   Column          Non-Null Count  Dtype  
---  ------          --------------  -----  
 0   Region          9 non-null      object 
 1   Anxiety_Score   9 non-null      float64
 2   Prevalence (%)  9 non-null      float64
dtypes: float64(2), object(1)
memory usage: 348.0+ bytes
None
```

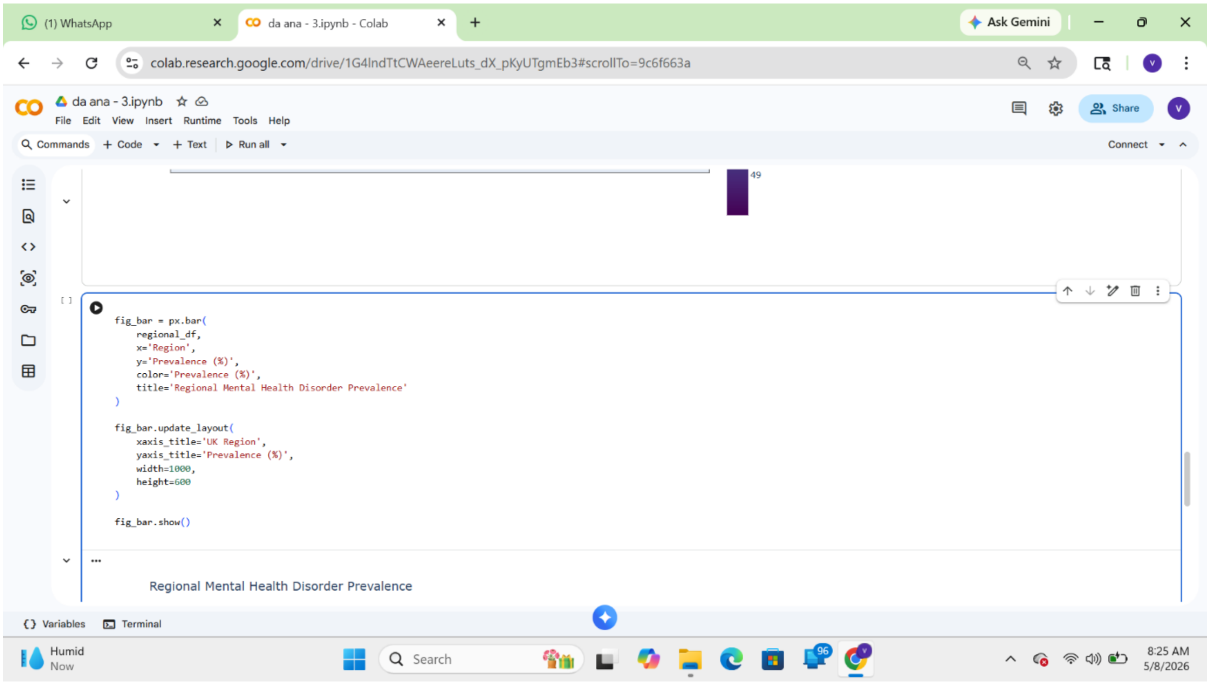


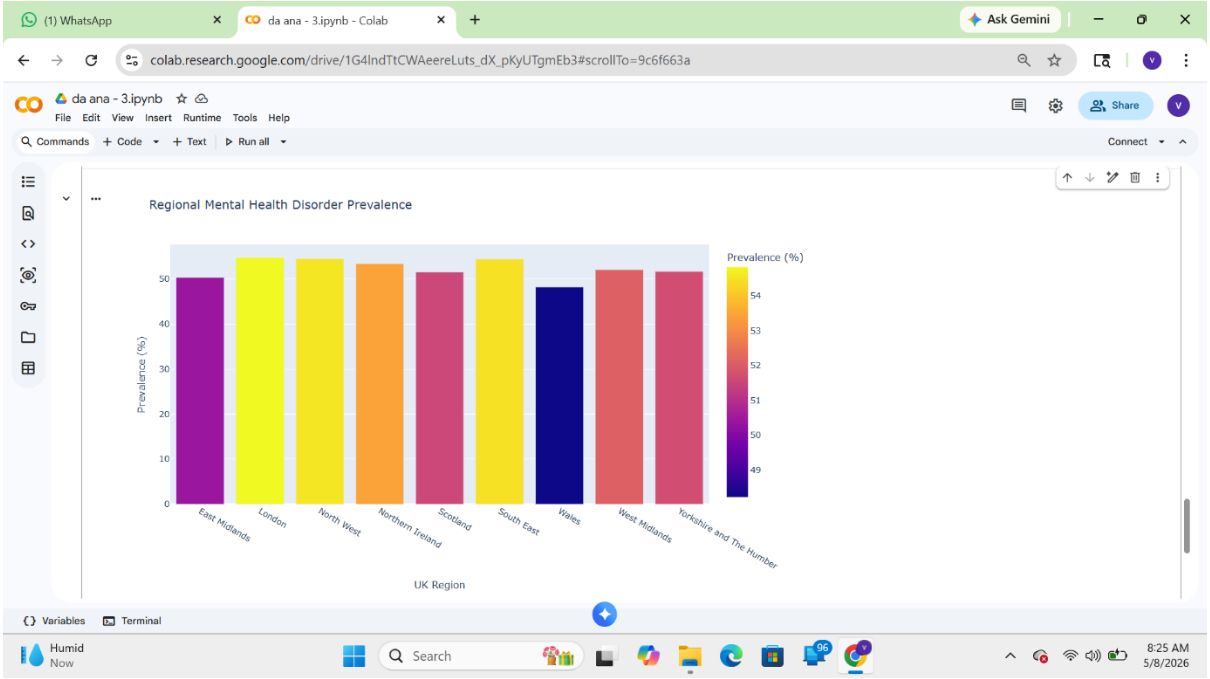
Choropleth Map





Regional Prevalence





Conclusion

The Regional Prevalence Mapping analysis successfully examined the distribution of mental health disorders such as anxiety and depression across different regions of the United Kingdom using the APMS-style dataset. The study demonstrated how geographic analysis can help identify regional variations in mental health prevalence and provide meaningful insights for healthcare planning and policy development.

Using Python, Pandas, Matplotlib, and Google Colab, the dataset was processed, grouped region-wise, and visualized through choropleth mapping techniques.

The choropleth map provided a clear visual understanding of areas with higher and lower prevalence rates.